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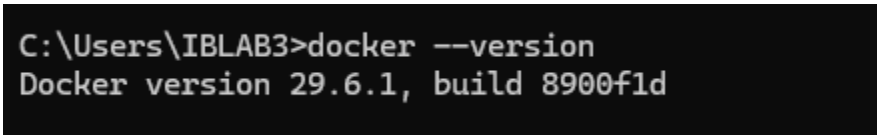
Docker - Container Management

Lab 2 : Docker Container Lifecycle and Management Using Nginx

Step 1: Check Docker Installation

docker --version



What is happening?

When you type:

docker --version

Docker checks whether the **Docker CLI (Command Line Interface)** is installed on your computer.

You
 │
 │ docker --version
 ▼
Docker CLI
 │
 ▼
Displays Version

Example Output

Docker version 28.2.2, build 123456

This command **does not contact Docker Hub or create anything**. It simply checks the Docker program installed on your machine.

Step 2: Search for Nginx

docker search nginx

What happens?

Docker connects to **Docker Hub**, which is the official online repository for Docker images.

Your Computer


Docker Hub

Returns Matching Images

Output

```
C:\Users\IBLAB3>docker search nginx
NAME                                DESCRIPTION                                     STARS     OFFICIAL
nginx                               Official build of Nginx.                        21337     [OK]
nginx/nginx-ingress                 NGINX and  NGINX Plus Ingress Controllers fo…  121
nginx/nginx-prometheus-exporter     NGINX Prometheus Exporter for NGINX and NGIN…  52
nginx/nginx-ingress-operator        NGINX Ingress Operator for NGINX and NGINX P…  4
nginx/nginxaas-loadbalancer-kubernetes                                              1
bitnamicharts/nginx                 Bitnami Helm chart for NGINX Open Source        4
ubuntu/nginx                        Nginx, a high-performance reverse proxy & we…  141
kasmweb/nginx                       An Nginx image based off nginx:alpine and in…  9
rancher/nginx                                                                       4
linuxserver/nginx                   An Nginx container, brought to you by LinuxS…  236
dtagdevsec/nginx                    T-Pot Nginx                                     0
paketobuildpacks/nginx                                                              0
vmware/nginx                                                                        3
gluufederation/nginx                 A customized NGINX image containing a consu…  1
cleanstart/nginx                    Secure by Design, Built for Speed, Hardened …  0
antrea/nginx                        Nginx server used for Antrea e2e testing        0
activestate/nginx                   ActiveState's customizable, low-to-no vulner…  0
intel/nginx                                                                         0
docksal/nginx                       Nginx service image for Docksal                 1
geokrety/nginx                      Our customized nginx image                      0
circleci/nginx                      This image is for internal use                  2
ilios/nginx                         Nginx customized to run Ilios along with the…  0
corpusops/nginx                     https://github.com/corpusops/docker-images/     1
wayofdev/nginx                      Nginx Docker image for PHP development. SSL-…  0
dockette/nginx                      Nginx SSL / HSTS / HTTP2                        3
```

Why?

Docker images are stored online. Before downloading one, you can search for available images.

Think of Docker Hub like the **Google Play Store** or **Apple App Store**, but instead of apps, it stores Docker images.

Step 3: Download Nginx

docker pull nginx

```
C:\Users\IBLAB3>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
8d2092eabbc5: Pull complete
062e450697fa: Pull complete
45ef33eb91c8: Pull complete
3b9f3509a826: Pull complete
6a24a6ebcf30: Pull complete
39694cde734e: Pull complete
0b19be717994: Pull complete
cd64a2895680: Download complete
29456554d205: Download complete
Digest: sha256:c90e4ea5905ee69515d84cd5520e11d204df06cce71a312666cebbc4b9b68267
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

What happens?

Docker downloads the **Nginx image** from Docker Hub to your computer.

Docker Hub
  |
  | Download Image
  ▼
Your Computer
  |
  ▼
Stored in Docker Images

After downloading:

docker images

shows

REPOSITORY TAG
nginx latest

Important

At this point:

✕No container exists.

Only the **image** exists.

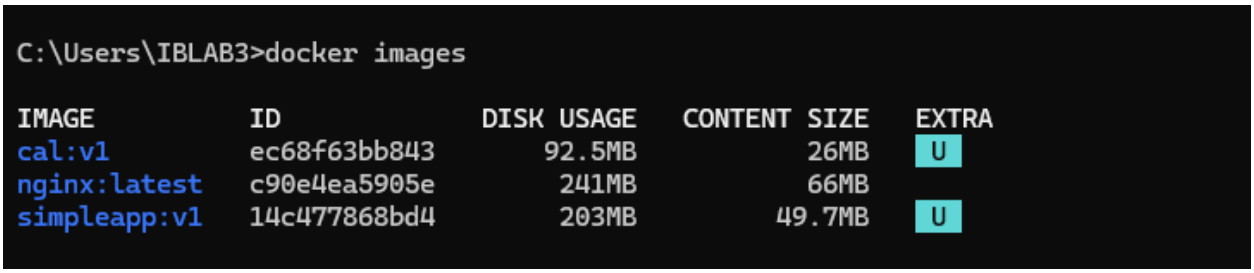
Think of it like:

Image = Software Installer

Container = Installed & Running Software

Step 4: View Images

docker images



What happens?

Docker shows all downloaded images.

Example

IMAGE ID
nginx
ubuntu
mysql

Nothing is running yet.

You only have installers.

Step 5: Run Nginx

docker run nginx

```
C:\Users\IBLAB3>docker run nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to p
erform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-d
efault.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/co
nf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/
conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envs
h
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templat
es.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-process
es.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 08:50:46 [notice] 1#1: using the "epoll" event method
2026/07/14 08:50:46 [notice] 1#1: nginx/1.31.2
2026/07/14 08:50:46 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 08:50:46 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 08:50:46 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 08:50:46 [notice] 1#1: start worker processes
2026/07/14 08:50:46 [notice] 1#1: start worker process 30
2026/07/14 08:50:46 [notice] 1#1: start worker process 31
2026/07/14 08:50:46 [notice] 1#1: start worker process 32
2026/07/14 08:50:46 [notice] 1#1: start worker process 33
2026/07/14 08:50:46 [notice] 1#1: start worker process 34
2026/07/14 08:50:46 [notice] 1#1: start worker process 35
2026/07/14 08:50:46 [notice] 1#1: start worker process 36
2026/07/14 08:50:46 [notice] 1#1: start worker process 37
2026/07/14 08:50:46 [notice] 1#1: start worker process 38
2026/07/14 08:50:46 [notice] 1#1: start worker process 39
2026/07/14 08:50:46 [notice] 1#1: start worker process 40
2026/07/14 08:50:46 [notice] 1#1: start worker process 41
2026/07/14 09:00:34 [notice] 1#1: signal 2 (SIGINT) received, exiting
2026/07/14 09:00:34 [notice] 31#31: exiting
2026/07/14 09:00:34 [notice] 32#32: exiting
```

This is one of the most important Docker commands.

What happens internally?

Docker performs several actions:

Step A

Checks whether the image exists.

Is nginx image available?

YES

If not:

Docker automatically performs:

docker pull nginx

Step B

Creates a **new container** from the image.

Image

↓

New Container

The image remains unchanged.

Containers are copies of the image.

Step C

Starts the Nginx server.

Container

↓

Nginx Starts

Now Nginx is running inside the container.

Step D

Your terminal becomes attached to the running container.

Terminal

↓

Container Output

Stopping the program using

Ctrl + C

stops the container.

Step 6: Detached Mode

docker run -d nginx

```
C:\Users\IBLAB3>
C:\Users\IBLAB3>docker run -d nginx
7a5bc3ef21adbdf62b0c138eec7fe0ae36632465c569a274de85eeefcf946af5
```

What changed?

-d

means

Detached Mode

Instead of connecting your terminal to the container:

Terminal

↓

Gets Prompt Back

The container continues running in the background.

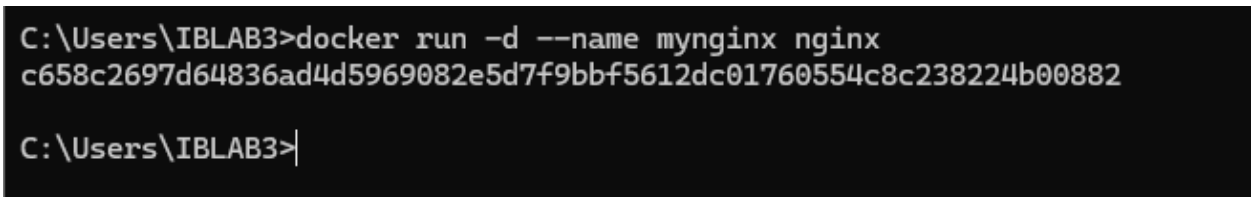
Docker returns

9f3ab2c89c8d...

This is the **Container ID**.

Step 7: Name the Container

docker run -d --name mynginx nginx



Without a name:

Docker generates random names.

Example

adoring_pike

crazy_turing

happy_einstein

Using

--name mynginx

makes management easier.

Instead of

docker stop 98ab123...

you simply type

docker stop mynginx

Step 8: Publish a Port

docker run -d --name webserver -p 8080:80 nginx

This is the most important networking concept.

```
C:\Users\IBLAB3>docker run -d --name webserver -p 3000:80 nginx
8cb89bbec4420545b62dd5e3bf37066943e65271166b7f5b839dd5010656d965
```

Inside the container

Nginx always listens on

Port 80
Container

Nginx

↓

80

But your computer cannot automatically access that internal port.

So Docker creates a mapping.

Computer

8080

↓

Container

80

Meaning

localhost:8080

↓

Container Port 80

Opening

http://localhost:8080

actually forwards the request to

Container

↓

Nginx

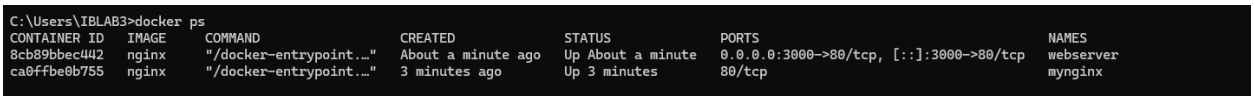
↓

Port 80

Step 9: List Running Containers

docker ps

Shows



Container ID

Image

Status

Ports

Name

Example

mynginx

Up 5 minutes

0.0.0.0:8080->80

Meaning

Host Port

8080

↓

Container Port

80

Step 10: Stop Container

docker stop mynginx

```
C:\Users\IBLAB3>docker stop mynginx
mynginx
```

Docker sends a signal to Nginx.

Please Stop Gracefully

Nginx saves its state and exits cleanly.

Container Status

Running

↓

Exited

Notice:

The **container is not deleted**.

It is only stopped.

Step 11: Start Container Again

docker start mynginx

```
C:\Users\IBLAB3>docker start mynginx
mynginx
```

Docker does **not create a new container**.

It simply starts the **existing** one.

Stopped Container

↓

Running Container

Step 12: Restart

docker restart mynginx

```
C:\Users\IBLAB3>docker restart mynginx
mynginx
```

Equivalent to

docker stop mynginx

docker start mynginx

Step 13: View Logs

docker logs mynginx

```
C:\Users\IBLAB3>docker logs mynginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/14 09:27:25 [notice] 1#1: using the "epoll" event method
2026/07/14 09:27:25 [notice] 1#1: nginx/1.31.2
2026/07/14 09:27:25 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/14 09:27:25 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/07/14 09:27:25 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/07/14 09:27:25 [notice] 1#1: start worker processes
2026/07/14 09:27:25 [notice] 1#1: start worker process 29
2026/07/14 09:27:25 [notice] 1#1: start worker process 30
2026/07/14 09:27:25 [notice] 1#1: start worker process 31
2026/07/14 09:27:25 [notice] 1#1: start worker process 32
2026/07/14 09:27:25 [notice] 1#1: start worker process 33
2026/07/14 09:27:25 [notice] 1#1: start worker process 34
2026/07/14 09:27:25 [notice] 1#1: start worker process 35
2026/07/14 09:27:25 [notice] 1#1: start worker process 36
2026/07/14 09:27:25 [notice] 1#1: start worker process 37
2026/07/14 09:27:25 [notice] 1#1: start worker process 38
2026/07/14 09:27:25 [notice] 1#1: start worker process 39
2026/07/14 09:27:25 [notice] 1#1: start worker process 40
2026/07/14 09:31:45 [notice] 1#1: signal 3 (SIGQUIT) received, shutting down
2026/07/14 09:31:45 [notice] 29#29: gracefully shutting down
2026/07/14 09:31:45 [notice] 30#30: gracefully shutting down
2026/07/14 09:31:45 [notice] 33#33: gracefully shutting down
2026/07/14 09:31:45 [notice] 34#34: gracefully shutting down
2026/07/14 09:31:45 [notice] 36#36: gracefully shutting down
2026/07/14 09:31:45 [notice] 29#29: exiting
2026/07/14 09:31:45 [notice] 37#37: gracefully shutting down
2026/07/14 09:31:45 [notice] 35#35: gracefully shutting down
2026/07/14 09:31:45 [notice] 38#38: gracefully shutting down
```

Every application writes messages.

Example

Nginx Started

Request Received

Response Sent

Docker stores these logs.

Useful for debugging.

Step 14: Enter the Container

docker exec -it mynginx bash

```
C:\Users\IBLAB3>docker exec -it mynginx bash
root@ca0ffbe0b755:/# docker rm mynginx
bash: docker: command not found
root@ca0ffbe0b755:/# exit
exit
```

This command opens a shell **inside the running container**.

Your Computer

↓

Docker

↓

Container

↓

Linux Bash

Now commands like

pwd
ls
cd /usr/share/nginx/html

are executed **inside the container**, not on your Windows machine.

This is very useful for troubleshooting or inspecting files.

Step 15: Remove the Container

docker rm mynginx

```
C:\Users\IBLAB3>docker rm mynginx
mynginx
```

Docker deletes the **container only**.

Container

✕Deleted

The image is still available.

You can create a new container anytime using:

docker run nginx

Step 16: Remove the Image

docker rmi nginx

```
C:\Users\IBLAB3>docker rmi nginx
Untagged: nginx:latest
Deleted: sha256:c90e4ea5905ee69515d84cd5520e11d204df06cce71a312666cebbc4b9b68267
```

Now Docker removes the **downloaded image** from your local system.

Image

✕Deleted

If any container still depends on that image, Docker will refuse to remove it until those containers are deleted.